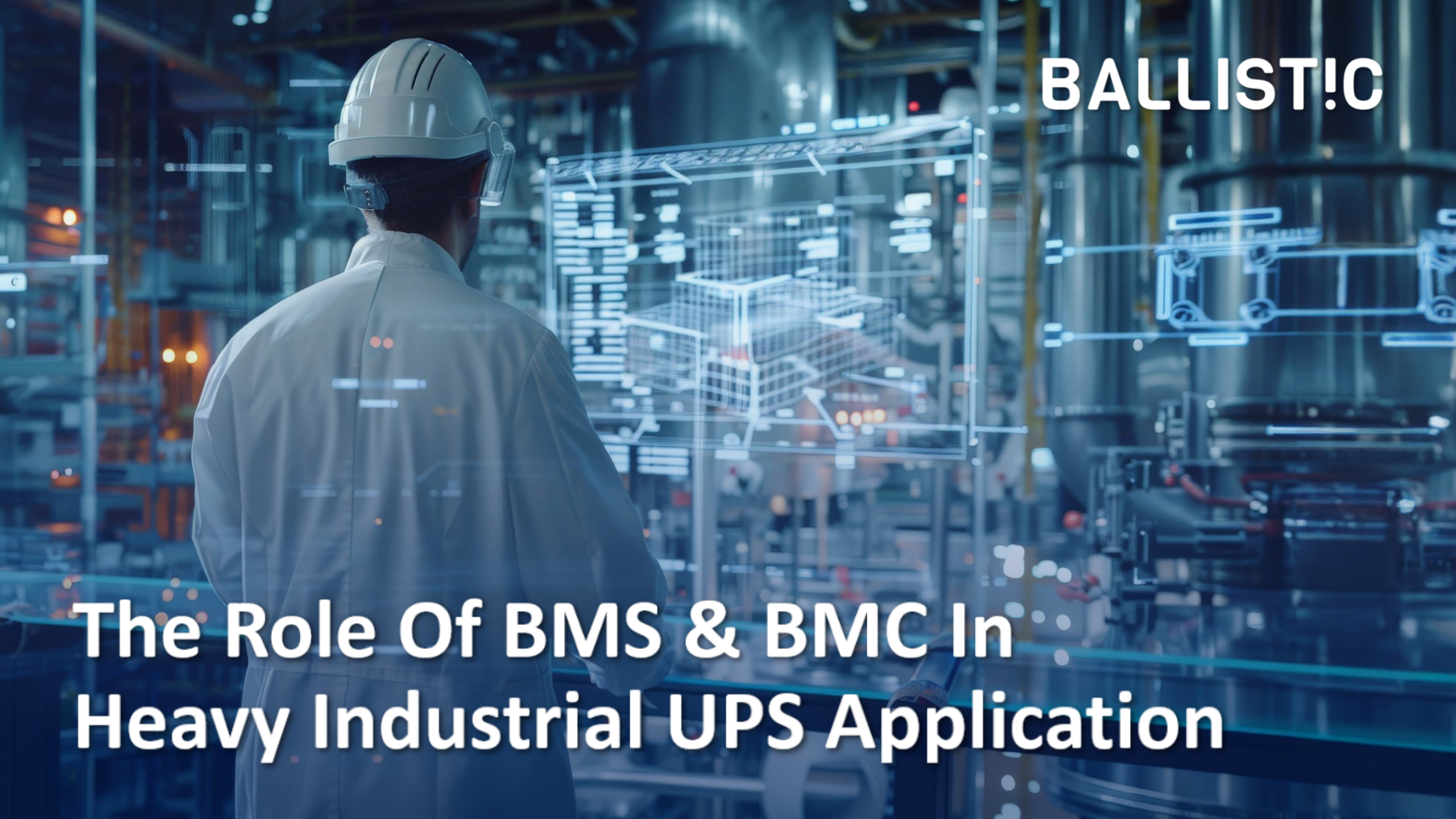




BALLIST!C

The Role Of BMS & BMC In Heavy Industrial UPS Application

Overview: BMC (Battery Monitoring Control)

- Back up batteries for electric companies and data centre. This power bank serves as a portable charger, providing electricity in areas without power outlets.
- Available in smaller size and giant size in form of mobile container.



Overview: Battery Management Systems (BMS)

- The BMS is the brain behind the UPS LFP battery pack.
- It ensures optimal performance, safety, and longevity by monitoring battery health.
- Prevents thermal runaway, a critical safety issue in heavy industrial environments.



Why is BMS Essential for Industrial UPS?

- High power demands and continuous operations require efficient battery management.
- BMS balances cells to maintain optimal charge and discharge cycles.
- Provides protection against short circuits, overheating, and overcurrent situations.

Key Functions of BMS for LFP UPS Systems

1. Cell Balancing - Ensures uniform charge among cells.
2. Voltage Monitoring - Prevents overcharging and deep discharging.
3. Current Measurement - Controls the flow of charging/discharging.
4. Temperature Regulation - Prevents thermal runaway and overheating.

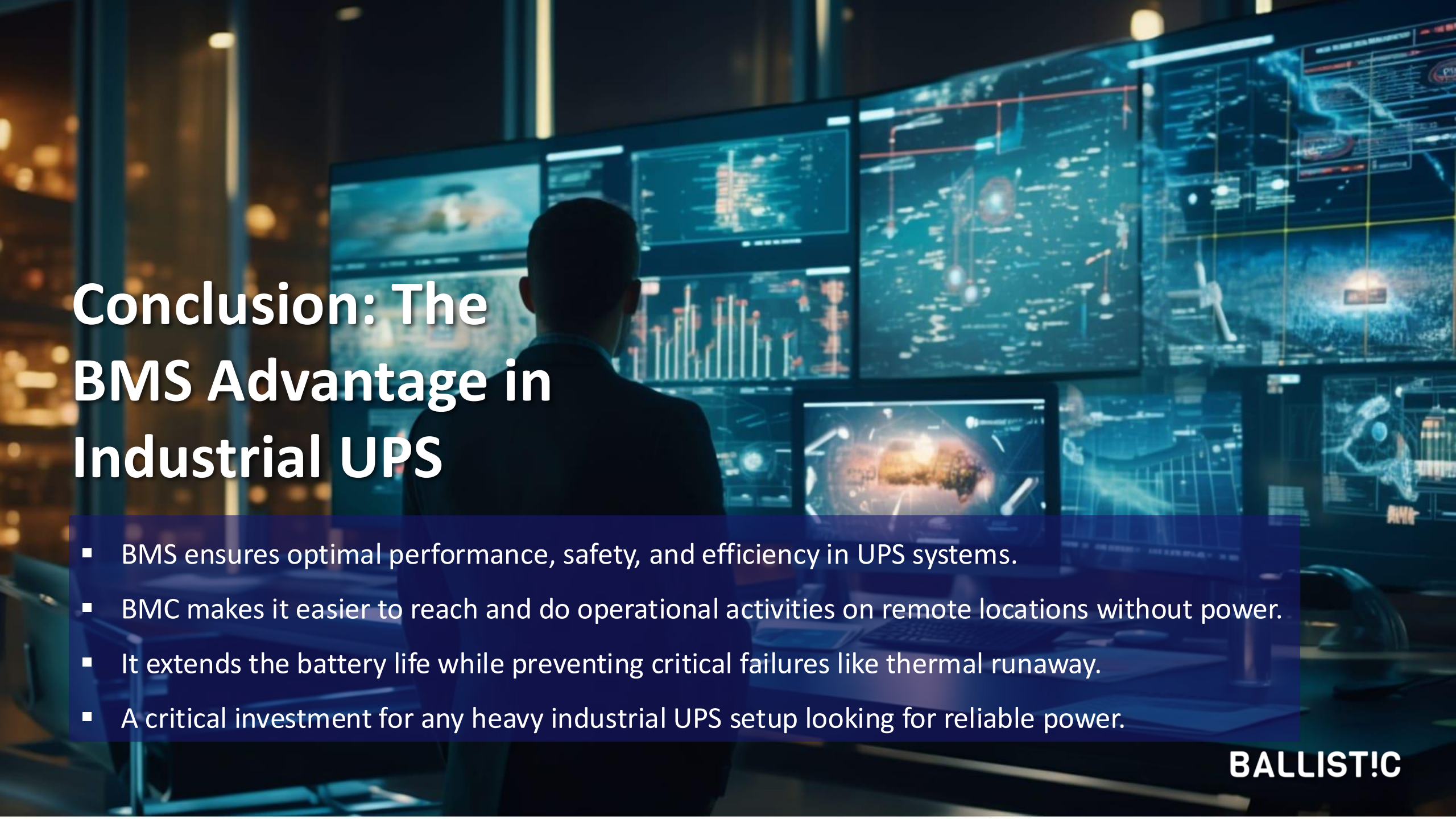
BMS Monitors & Maintains Battery Health

- State of Charge (SOC): Real-time capacity data for power management.
- State of Health (SOH): Evaluates overall health and longevity.
- Predictive Maintenance: BMS insights lead to better scheduling of replacements.

Current Trends in BMS for UPS LFP Batteries

- AI & Cloud-based Monitoring: Predictive analytics for battery life & performance.
- Scalability: BMS designed for modular UPS systems, adaptable to power needs.
- Enhanced Safety Protocols: Modern BMS systems with advanced fault detection.





Conclusion: The BMS Advantage in Industrial UPS

- BMS ensures optimal performance, safety, and efficiency in UPS systems.
- BMC makes it easier to reach and do operational activities on remote locations without power.
- It extends the battery life while preventing critical failures like thermal runaway.
- A critical investment for any heavy industrial UPS setup looking for reliable power.